Exercise 2.11. Prove that if m, n and t are integers, then at least one of $m-n$, $n-t$ and $m-t$ is even.

Solution.

We need to prove that for any integers m, n, t, **at least one** of

$$m-n, \qquad n-t, \qquad m-t$$

is even.

We use the fact that every integer is either **even or odd.**

There are two cases.

Case 1: m and n have the same parity

If m and n are both even or both odd, then their difference is even. Therefore,

$$\boxed{m-n \text{ is even}}.$$

So the statement is true.

Case 2: m and n have different parity

Suppose m is even and n is odd (the reverse situation works the same way).

Now consider t.

- If t is even, then m and t have the same parity, so

$$\boxed{m-t \text{ is even}}.$$

- If t is odd, then n and t have the same parity, so

$$\boxed{n-t \text{ is even}}.$$

Thus, in every possible case, at least one of

$$m-n, \quad n-t, \quad m-t$$

is even.

Therefore,

$$\boxed{\text{at least one of } m-n,\ n-t,\ m-t \text{ is even.}}$$